```
/usr/local/bin/python3.9 "/Users/rohitchavan/Library/Application Support/JetBrains/PyCharmCE2020.3/scratches/scratch_1.py"
Maximum profit: 240.0
  profit=60  weight=10  fraction taken=1.00
  profit=100 weight=20  fraction taken=1.00
  profit=120 weight=30  fraction taken=0.67

Process finished with exit code 0
```

payment.py | scratch_1.py

```python
def fractional_knapsack(profits, weights, capacity):
    n = len(profits)

    items = [(profits[i], weights[i], profits[i] / weights[i])
             for i in range(n)]

    items.sort(key=lambda item: item[2], reverse=True)

    total_profit = 0.0
    remaining = capacity
    chosen = []

    for profit, weight, ratio in items:
        if remaining <= 0:
            break

        if weight <= remaining:
            total_profit += profit
            remaining -= weight
            chosen.append((profit, weight, 1.0))
        else:
            fraction = remaining / weight
            total_profit += profit * fraction
            chosen.append((profit, weight, fraction))
            remaining = 0

    return total_profit, chosen
profits = [60, 100, 120]
weights = [10, 20, 30]
capacity = 50

total, chosen = fractional_knapsack(profits, weights, capacity)
print("Maximum profit:", total)
for p, w, frac in chosen:
    print(f"  profit={p:<3} weight={w:<3} fraction taken={frac:.2f}")
```

for p, w, frac in chosen